

Theme 5: Integration

Note Title

5/2/2018

Unit 5.1 Anti-derivatives

If $F' = f$, then F is the anti-derivative of f .

Reverse process of differentiation.

We also write:

$$F = \int f(x) dx \quad \text{or} \quad f = \int f'(x) dx$$

Examples:

Always test your answer!!

1) If $f'(x) = \sin x$, then $f(x) =$

2) $f'(x) = x^2 + x$, then $f(x) =$

3) $f'(x) = 3x^2 \cos(x^3)$, then $f(x) =$

$f'(x) = x^2 \cos(x^3)$, then $f(x) =$

Find the anti-derivative F of

1) $f(x) = e^x \sin e^x$

2) $f(x) = x(x^2+1)^3$

3) $f(x) = x \sec^2 x^2$

Always test
your answer!!

SOLUTIONS.

① If $f'(x) = \sin x$, then $f(x) = -\cos x + k$, because:

TEST: $\frac{d}{dx}(-\cos x + k) = -(-\sin x) + 0 = \sin x$.

② $f'(x) = x^2 + x$, then $f(x) = \frac{1}{3}x^3 + \frac{1}{2}x^2 + k$.

TEST: $\frac{d}{dx}(\frac{1}{3}x^3 + \frac{1}{2}x^2 + k) = x^2 + x + 0$

③ If $f'(x) = 3x^2 \cos(x^3)$, then $f(x) = \sin(x^3) + k$

TEST: $\frac{d}{dx}(\sin x^3 + k) = (\cos x^3) \times 3x^2 + 0$

If $f'(x) = x^2 \cos(x^3)$, then $f(x) = \frac{1}{3} \sin x^3 + k$

TEST: $\frac{d}{dx}(\frac{1}{3} \sin(x^3) + k) = \frac{1}{3} \cos x^3 \times 3x^2 + 0$
 $= x^2 (\cos x^3) + 0$

Find the anti-derivative F of f :

① $f(x) = e^x \sin e^x$

$\therefore$ Which function F must I differentiate to get $f(x)$?

$$F(x) = -\cos e^x + k$$

$$\begin{aligned}\text{Test: } \frac{d}{dx} F(x) &= \frac{d}{dx} (-\cos e^x + k) \\ &= -(-\sin e^x) \times e^x + 0 \\ &= \underline{e^x \sin e^x}\end{aligned}$$

② $f(x) = x(x^2+1)^4$

What function must I differentiate to obtain $f(x)$?

$$F(x) = \frac{1}{10} (x^2+1)^5 + k.$$

$$\begin{aligned}\text{Test: } \frac{d}{dx} (F(x)) &= \frac{d}{dx} \left(\frac{(x^2+1)^5}{10} + k \right) \\ &= \frac{5(x^2+1)^4 \times 2x}{10} + 0 \\ &= x(x^2+1)^4\end{aligned}$$

③ $f(x) = x \sec^2 x^2$

What is $F(x)$ if $\frac{dF}{dx} = x \sec^2 x^2$?

$$F(x) = \frac{1}{2} \tan x^2 + k$$

$$\begin{aligned}\text{Test: } \frac{d}{dx} F(x) &= \frac{d}{dx} \left(\frac{1}{2} \tan x^2 + k \right) \\ &= \frac{1}{2} \sec^2 x^2 \times 2x + 0 \\ &= \underline{x \sec^2 x^2}\end{aligned}$$